

North Sydney Girls High School.  
2nd Trial. 1989.

QUESTION 1:

- (a) Find correct to three decimal places the value of:

$$\frac{5.6 \times 4.9^3}{\sqrt{7.3} + 4.1}$$

- (b) Solve the equation:

$$x - \frac{x+1}{3} = 2 + \frac{x}{5}$$

- (c) Last year Council rates increased by  $7\frac{1}{2}\%$ .  
The new rate for a property is \$1735.  
What was the old rate for this property?  
Give your answer correct to the nearest dollar.
- (d) The point P (0, -4) is the mid-point of A (g, -2) and B (-5, h).  
Find the values of g and h.
- (e) A carton contains a dozen eggs, 3 of which have a double yolk.  
If 3 eggs are required to make a cake, find the probability  
that at least one of the eggs used is a double yolk.

QUESTION 2:

(a) Determine:  $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

(b) Differentiate the following with respect to x:

(i)  $\frac{1}{\sqrt{x}}$

(ii)  $x^2 e^{-x}$

(iii)  $(\cos x + \sin x)^3$

(iv)  $\log \left( \frac{x+2}{x-2} \right)$

(c) If  $\alpha$  and  $\beta$  are the roots of the equation

$$2x^2 - 4x + 1 = 0 \text{ find:}$$

(i)  $\alpha + \beta$

(ii)  $\alpha\beta$

(iii)  $\alpha^2 + \alpha\beta + \beta^2$

(d) Find the area of the sector of a circle in which the arc length is 15 cm and the radius is 7 cm.

QUESTION 3:

(a) Find the primitive function of each of the following:

(i)  $12x^2 - 4$

(ii)  $e^{7x} + 14$

(iii)  $(3x + 5)^6$

(b) Evaluate:  $\int_1^{13} \frac{dx}{2x + 6}$

(c) Find all values of  $k$  for which the quadratic equation  $kx^2 - 8x + k = 0$  has real roots.

(d) The gradient function of a curve is given by

$$\frac{dy}{dx} = 3 - 4x$$

Find the equation of the curve if it passes through the point  $(6, -5)$ .

QUESTION 4

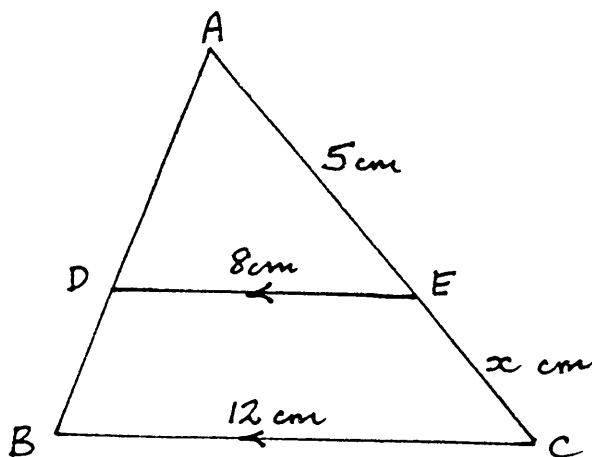
- (a) P is the point  $(-2, 7)$   
d is the line  $2x + 5y + 1 = 0$   
k is the line through P, perpendicular to d.
- (i) Find the equation of k.
- (ii) If d meets the y-axis at D, and k meets the y-axis at K, find the area of the triangle PKD.
- (b) Solve:  $m^4 - 6m^2 - 40 = 0$
- (c) Find the equation of the circle which is concentric with the circle  $x^2 + y^2 + 8x + 2y + 8 = 0$  and which passes through the point  $(1, 7)$ .
- (d) Find the values of x for which  $|2x - 1| \leq 5$

QUESTION 5

- (a) The fifth term of an arithmetic series is 14,  
and the sum of the first 10 terms is 165.  
Find the first term of this series.

- (b) Find the size of each internal angle of a  
regular pentagon. Hence or otherwise find  
the size of each of the external angles.

(c)



Find  $x$ , giving reasons in full.

(Diagram not drawn to scale)

- (d) PQRS is a quadrilateral. Its diagonals are perpendicular,  
meeting at T, which is the midpoint of QS, but not of PR.
- (i) Draw a diagram showing this information.
- (ii) Prove that  $\hat{PQR} = \hat{PSR}$

QUESTION 6

- (a) If  $\cos \theta^\circ = -\frac{5}{11}$  and  $180^\circ < \theta < 360^\circ$

give the exact value of  $\operatorname{cosec} \theta^\circ$

- (b) From a lighthouse, L, a ship, S, bears  $053^\circ$  T and is at a distance of 8 nautical miles.  
From L a boat B bears  $293^\circ$  T and is a distance of 6 nautical miles.

(i) Draw a diagram marking on it the information supplied.

(ii) Find the distance of Ship S from boat B.  
Give your answer as a surd.

(iii) Find the bearing of ship S from boat B.  
Give your answer to the nearest degree.

- (c) Tennis balls are often supplied in a cylindrical container which holds 3 balls in a neat fit (the centre of the balls are collinear.) What percentage of the volume of the container is occupied by the balls?

QUESTION 7

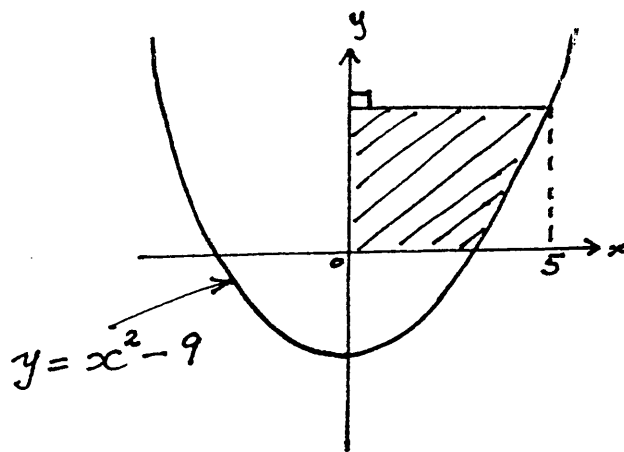
- (a) If  $S_n = n(n - 2)$ , find an expression for the  $n$ th term of the series.
- (b) Which term of 2, 6, 18, ... is 486?
- (c) A person invests \$500 at the beginning of each year in a superannuation fund.  
Compound interest is paid at 10% per annum on the investment.  
The first \$500 is to be invested at the beginning of 1989  
and the last is to be invested at the beginning of 2018.  
Calculate, to the nearest dollar:
- (i) the amount to which the 1989 investment will have grown by the beginning of 2019.
  - (ii) the amount to which the total investment will have grown by the beginning of 2019.
- (d) Solve  $2x^2 + 4x - 7 = 0$  in exact form.

QUESTION 8

(a) For the parabola  $x^2 - 8y = 0$

- (i) find the coordinates of the Vertex
- (ii) find the coordinates of the Focus
- (iii) find the equation of the Directrix
- (iv) find the equation of the tangent to the parabola at the point where  $x = 4$

(b) If the shaded region is rotated around the y axis, find the volume of the solid of revolution generated.



(c) Differentiate  $e^{x^2}$ . Hence or otherwise

find  $\int_0^1 x e^{x^2} dx$ .

QUESTION 9

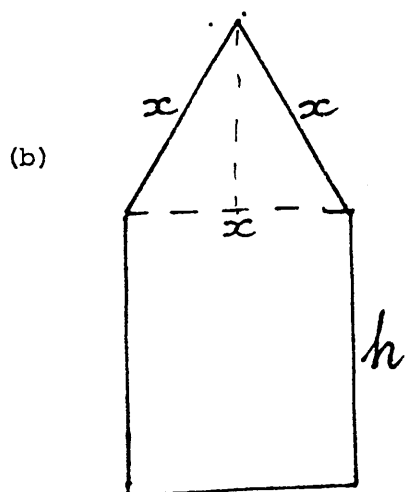
- (a) Sketch the curve  $y = e^{-x} + 2$ . Evaluate the area bounded by the curve, the  $x$  axis and the ordinates  $x = 0$  and  $x = 3$ .
- (b) For the curve  $y = x^3 + 3x^2 - 9x$
- (i) Find the stationary points and determine their nature.
  - (ii) Find any points of inflection.
  - (iii) Using the above information and any other relevant information, sketch the curve  $y = x^3 + 3x^2 - 9x$ .
  - (iv) Find the set of values for which the curve is monotonically increasing.

QUESTION 10

- (a) The following table gives values of  $f(x) = x \log x$

|      |   |      |      |      |      |
|------|---|------|------|------|------|
| x    | 1 | 2    | 3    | 4    | 5    |
| f(x) | 0 | 1.39 | 3.30 | 5.55 | 8.05 |

Use Simpson's Rule with five function values to find an approximation for the value of  $\int_1^5 x \log x \, dx$



The figure represents a large window made up of a rectangle and an equilateral triangle.

The sides of the equilateral triangle are  $x$  m and the dimensions of the rectangle are  $x$  m and  $h$  m as shown.

- (i) If the beading around the outside of the window is 16 m long show that  $h = 8 - \frac{3}{2}x$ .
- (ii) Show that an expression for the Area of the window is given by
- $$A = 8x + \frac{x^2}{4} (\sqrt{3} - 6)$$
- (iii) Hence calculate the dimensions of the window which allow the maximum light to pass through (correct to 2 decimal places).

1) (a) 19.768 (1)

(b)  $x - \frac{(x+1)}{3} = 2 + \frac{x}{5}$

$\therefore 15x - 5(x+1) = 30 + 3x$  ← 1 mark

$\therefore 15x - 5x - 5 = 30 + 3x$

$10x - 5 = 30 + 3x$

$\therefore 7x = 35$

$\therefore \underline{x=5}$  (2)

(c) Old rate 100%

New rate 107½% is \$1735

$\therefore 1\% \text{ is } \$ \frac{1735}{107.5}$

$\therefore 100\% \text{ is } \$ \frac{1735}{107.5} \times 100$

$= \$1614$  (2)

(to nearest dollar)

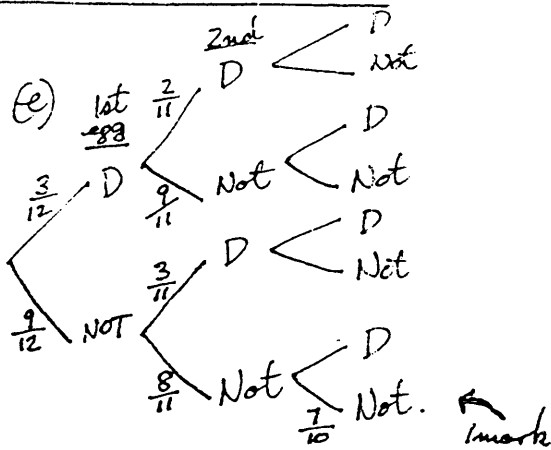
(d)  $0 = \frac{g-5}{2} \quad -4\frac{1}{2} = \frac{-2+h}{2}$

$\therefore 0 = g-5 \quad \therefore -\frac{9}{2} = \frac{-2+h}{2}$

$\therefore \underline{g=5} \quad \therefore -9 = -2+h$

← 1 each  $\therefore \underline{h=-7}$

(2)



P(at least 1 D) ← 1 mark

$= 1 - P(\text{No double-yolies})$

$= 1 - P(\text{Not} \times \text{Not} \times \text{Not})$

$= 1 - \frac{9}{12} \times \frac{8}{11} \times \frac{7}{10}$

$= 1 - \frac{21}{55}$

$= \frac{34}{55}$

(3)

10

2) (a)  $\lim_{x \rightarrow 3} \frac{x^2-9}{x-3}$

$= \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{(x-3)}$

$= \lim_{x \rightarrow 3} (x+3)$

$= \underline{6}$

(1)

2.

$$b)(i) y = x^{-\frac{1}{2}}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2} x^{-\frac{3}{2}}$$

$$= \frac{-1}{2x^{\frac{3}{2}}} \quad (1)$$

$$(ii) y = x^2 e^{-x}$$

$$\begin{aligned} \frac{dy}{dx} &= (x^2) \cdot (e^{-x} \cdot -1) + (e^{-x}) \cdot (2x) \\ &= \underline{x e^{-x} (2-x)} \quad (1) \end{aligned}$$

$$(iii) y = (\cos x + \sin x)^3$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= 3(\cos x + \sin x)^2 (-\sin x + \cos x) \\ &= 3(\cos x + \sin x)^2 (\cos x - \sin x) \quad (1) \end{aligned}$$

$$(iv) y = \log \left( \frac{x+2}{x-2} \right)$$

$$y = \log(x+2) - \log(x-2)$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{1}{x+2} - \frac{1}{x-2} \\ &= \frac{(x-2) - (x+2)}{x^2-4} \\ &= \frac{x-2-x-2}{x^2-4} \\ &= \frac{-4}{x^2-4} \quad (1) \end{aligned}$$

$$(c) 2x^2 - 4x + 1 = 0$$

$$\begin{aligned} (i) \alpha + \beta &= -\frac{b}{a} \\ &= -\frac{(-4)}{2} \\ &= \underline{2} \quad (1) \end{aligned}$$

$$(ii) \alpha\beta = \frac{c}{a}$$

$$= \frac{1}{2} \quad (1)$$

$$(iii) \alpha^2 + \alpha\beta + \beta^2$$

$$= (\alpha + \beta)^2 - \alpha\beta$$

$$= (2)^2 - \left(\frac{1}{2}\right)$$

$$= 4 - \frac{1}{2} \quad (1)$$

$$= \underline{3\frac{1}{2}}$$

$$(d) l = r\theta$$

$$\therefore \theta = \frac{l}{r}$$

$$= \frac{15}{7} \leftarrow \text{mark.}$$

$$A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 49 \times \frac{15}{7}$$

$$= \frac{7 \times 15}{2}$$

$$= \frac{105}{2}$$

$$= \underline{52\frac{1}{2} \text{ cm}^2} \quad (2)$$

[10]

3) (a)

$$\begin{aligned} \text{(i)} \quad \int 12x^2 - 4 \, dx \\ = \frac{12x^3}{3} - 4x + C \\ = 4x^3 - 4x + C \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \int e^{7x} + 14 \, dx \\ = \frac{e^{7x}}{7} + 14x + C \quad \text{①} \end{aligned}$$

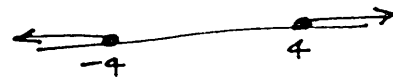
$$\begin{aligned} \text{(iii)} \quad \int (3x+5)^6 \, dx \\ = \frac{(3x+5)^7}{7 \cdot 3} + C \\ = \frac{(3x+5)^7}{21} + C \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \int_1^{13} \frac{dx}{2x+6} \\ = \frac{1}{2} \left[ \log(2x+6) \right]_1^{13} \quad \leftarrow \text{mark.} \\ = \frac{1}{2} \{ \log 45 - \log 3 \} \\ = \frac{1}{2} \log \frac{45}{3} \\ = \frac{1}{2} \log 15 \quad (\text{or } \log \sqrt{15}) \quad \text{②} \end{aligned}$$

(c) For real roots  $\Delta \geq 0$  ↖ mark

$$\begin{aligned} \Delta &= b^2 - 4ac \\ &= (-8)^2 - 4(k)(k) \\ &= 64 - 4k^2 \end{aligned}$$

$$\begin{aligned} \therefore 4k^2 - 64 &\leq 0 \\ \therefore 4(k^2 - 16) &\leq 0 \quad \leftarrow \text{mark} \\ \therefore (k-4)(k+4) &\geq 0 \end{aligned}$$



$$\therefore k \leq -4, \quad k \geq 4 \quad \text{③}$$

$$\begin{aligned} \text{(d)} \quad \frac{dy}{dx} &= 3 - 4x \\ \therefore y &= 3x - \frac{4x^2}{2} + C \\ \therefore y &= 3x - 2x^2 + C \quad \leftarrow \text{mark.} \\ \text{Thru' } (6, -5) \\ -5 &= 18 - 72 + C. \\ -5 &= -54 + C \\ 49 &= C \end{aligned}$$

$$\therefore y = -2x^2 + 3x + 49 \quad \text{②} \quad \boxed{10}$$

4) (a) d is  $2x + 5y + 1 = 0$   
 (i)  $\therefore 5y = -2x - 1$   
 $\therefore y = -\frac{2}{5}x - \frac{1}{5}$

$\therefore$  slope of k is  $\frac{5}{2}$  ← 1 mark

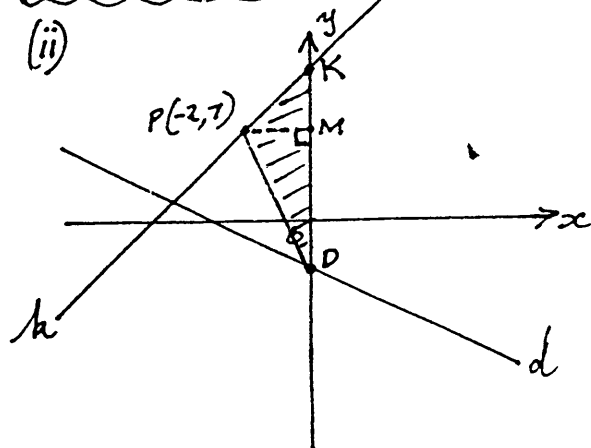
Thru'  $(-2, 7)$  eqn of k is

$$(y - 7) = \frac{5}{2}(x + 2)$$

$$\therefore 2(y - 7) = 5(x + 2)$$

$$2y - 14 = 5x + 10 \quad (2)$$

$$\therefore 5x - 2y + 24 = 0 \quad (\text{or } y = \frac{5}{2}x + 12)$$



D is point  $(0, -\frac{1}{5})$

K is point  $(0, 12)$

$$\therefore \text{length } KD = 12 + \frac{1}{5}$$

$$= 12\frac{1}{5}$$

From sketch PM is 2 } 1 mark

Area  $\Delta PKD = \frac{1}{2} \text{ base} \times \text{height}$

$$= \frac{1}{2} \times KD \times PM$$

$$= \frac{1}{2} \times 12\frac{1}{5} \times 2$$

$$= \frac{1}{2} \times \frac{61}{5} \times 2 \quad (2)$$

$$= 12\frac{1}{5} \text{ units}^2$$

4.  
 (b)  $m^4 - 6m^2 - 40 = 0$

Let  $x = m^2$

$$\therefore x^2 - 6x - 40 = 0$$

$$(x + 4)(x - 10) = 0 \quad \leftarrow 1 \text{ mark}$$

$$\therefore x = -4, 10$$

$$\therefore m^2 = -4 \text{ or } m^2 = 10$$

$$\text{NO SOLN.} \quad \therefore m = \pm\sqrt{10} \quad (2)$$

(c)  $x^2 + y^2 + 8x + 2y + 8 = 0$

$$\therefore x^2 + 8x + 16 + y^2 + 2y + 1 = -8 + 16 + 1$$

$$\therefore (x + 4)^2 + (y + 1)^2 = 9 \quad \leftarrow 1 \text{ mark.}$$

New circle also has centre

$(-4, -1)$ .  $\therefore$  New circle is

$$(x + 4)^2 + (y + 1)^2 = r^2$$

Thru'  $(1, 7)$

$$\therefore (1 + 4)^2 + (7 + 1)^2 = r^2$$

$$\therefore 25 + 64 = r^2$$

$$89 = r^2$$

$\therefore$  Circle is

$$(x + 4)^2 + (y + 1)^2 = 89. \quad (2)$$

(d)

$$2x - 1 \leq 5 \text{ or } -(2x - 1) \leq 5 \quad \leftarrow 1 \text{ mark}$$

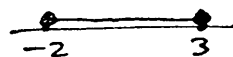
$$\therefore 2x \leq 6$$

$$x \leq 3$$

$$-2x + 1 \leq 5$$

$$-2x \leq 4$$

$$x \geq -2$$



$$\therefore -2 \leq x \leq 3 \quad (2)$$

[10]

5) (a)  $T_n = a + (n-1)d$   
 $\therefore 14 = a + 4d \quad \text{--- (1)}$   
 $S_n = \frac{n}{2} \{2a + (n-1)d\}$   
 $\therefore 165 = \frac{10}{2} \{2a + 9d\}$   
 $165 = 5(2a + 9d)$   
 $33 = 2a + 9d \quad \text{--- (2)}$

(1)  $\times 9 \quad 126 = 9a + 36d \quad \text{--- (3)}$

(2)  $\times 4 \quad 132 = 8a + 36d \quad \text{--- (4)}$

(3) - (4)  $-6 = a$

$\therefore$  First term is -6 (2)

(b) Angle Sum (S) =  $(2n-4)$  rt angles

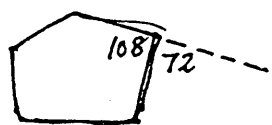
$\therefore S = (10-4) \times 90$

$S = 6 \times 90$

$S = 540$

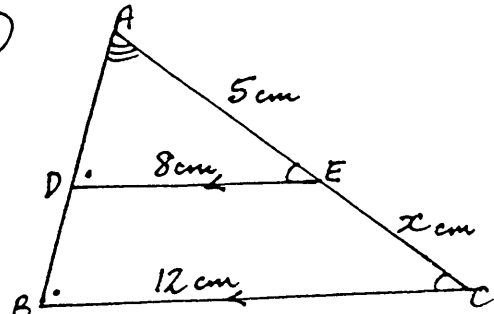
$\therefore$  Each angle is  $\frac{540}{5}$

$= 108^\circ$  (mark)



$\therefore$  each external angle is  $72^\circ$  (OR  $\frac{360}{5} = 72^\circ$ ) (2)

(c)



$\hat{ADE} = \hat{ABC}$  (corr,  $DE \parallel BC$ )

$\hat{AED} = \hat{ACB}$  (corr,  $DE \parallel BC$ )

$\hat{DAE}$  is common

$\therefore \triangle ADE \equiv \triangle ABC$  (equiangular) (mark)

$\therefore \frac{8}{12} = \frac{5}{(x+5)}$  (corr sides in sim.  $\Delta$ 's)

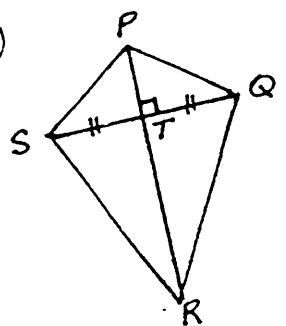
$\therefore 8(x+5) = 60$

$8x + 40 = 60$

$\therefore 8x = 20$

$\therefore x = 2\frac{1}{2}$  (2)

(d) (i)



(2)

(ii) In  $\Delta$ 's PTS, PTQ

$ST = TQ$  (data)

PT is common

$\hat{PTQ} = \hat{PTS}$  ( $= 90^\circ$  data)

$\therefore \triangle PTS \equiv \triangle PTQ$  (SAS)

$\therefore \hat{PST} = \hat{PQT}$  (corr  $\angle$ 's in  $\equiv \Delta$ 's)

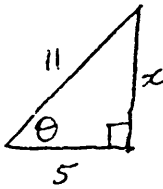
Similarly  $\hat{TSR} = \hat{TQR}$  (mark)

But  $\hat{PQR} = \hat{PQT} + \hat{TQR}$

and  $\hat{PSR} = \hat{PST} + \hat{TSR}$

$\therefore \hat{PQR} = \hat{PSR}$  (2)

6) (a)  $\cos \theta = -\frac{5}{11}$



$$11^2 = x^2 + 5^2$$

$$121 = x^2 + 25$$

$$x = \sqrt{96} \leftarrow \text{1 mark}$$

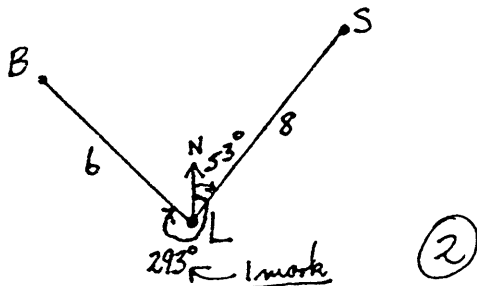
S | A  
— | —  
T | C

3rd quad since  
 $180 < \theta < 360$

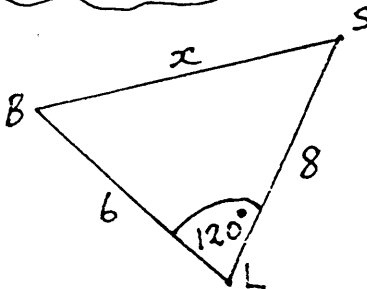
$$\therefore \operatorname{cosec} \theta = -\frac{11}{\sqrt{96}} \quad (2)$$

$$\left( \text{or } -\frac{11}{4\sqrt{6}} \text{ or } -\frac{11\sqrt{6}}{24} \right)$$

(b)(i)



(ii)



$$x^2 = 6^2 + 8^2 - 2 \cdot 6 \cdot 8 \cdot \cos 120$$

$$= 36 + 64 - 96 \cos (180 - 60)$$

$$= 100 + 96 \cos 60 \quad \begin{matrix} S/A \\ T/C \end{matrix}$$

$$= 100 + 96 \cdot \frac{1}{2}$$

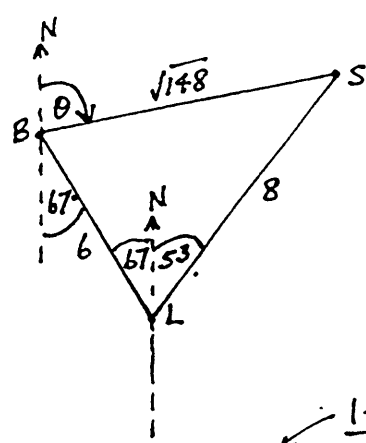
$$= 100 + 48$$

$$= 148$$

$$\therefore x = \sqrt{148}$$

$$\therefore \text{dist SB is } \underline{\underline{\sqrt{148}}} \text{ (or } 2\sqrt{37}) \text{ n.miles.} \quad (2)$$

(iii)



$$\frac{\sin \widehat{SBL}}{8} = \frac{\sin 120}{\sqrt{148}}$$

$$\therefore \sin \widehat{SBL} = \frac{8 \sin 120}{\sqrt{148}}$$

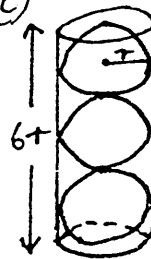
$$\therefore \widehat{SBL} = 35^\circ$$

$$\therefore \theta = 180 - (67 + 35)$$

$$= 78^\circ$$

$$\therefore S \text{ is } \underline{\underline{078^\circ T}} \text{ from B.} \quad (2)$$

(c)



$$V \text{ of balls} = 3 \times \frac{4}{3} \pi r^3$$

$$= 4\pi r^3$$

$$V \text{ of cylinder} = \pi r^2 h$$

$$= \pi r^2 \times 6r$$

$$= 6\pi r^3$$

$$\therefore \% \text{ occupied} = \frac{4\pi r^3}{6\pi r^3} \times 100 \%$$

(mark both volumes).

$$= \frac{200}{3} \%$$

$$= \underline{\underline{66\frac{2}{3} \%}} \quad (2)$$

10

$$\begin{aligned}
 7) (a) T_n &= S_n - S_{n-1} \quad \text{1 mark} \\
 &= [n(n-2)] - [(n-1)(n-3)] \\
 &= (n^2 - 2n) - (n^2 - 4n + 3) \\
 &= n^2 - 2n - n^2 + 4n - 3 \\
 &= \underline{2n - 3} \quad (2)
 \end{aligned}$$

(b) 2, 6, 18, --- is a G.P.

$$a = 2 \quad r = 3 \quad T_n = ar^{n-1}$$

$$\begin{aligned}
 \therefore 486 &= (2)(3)^{n-1} \quad \text{1 mark} \\
 243 &= 3^{n-1} \quad \text{OR } 3^5 = 3^{n-1}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \log 243 &= \log 3^{(n-1)} \quad \left\{ \begin{array}{l} \therefore 5 = n-1 \\ \therefore 6 = n \end{array} \right. \\
 \therefore \log 243 &= (n-1) \log 3 \\
 \therefore n-1 &= \frac{\log 243}{\log 3}
 \end{aligned}$$

$$\therefore n-1 = 5 \quad (2)$$

$$\therefore \underline{n = 6}$$

(c) (i)

$$\begin{aligned}
 A &= P \left( 1 + \frac{r}{100} \right)^n \\
 &= 500 \left( 1 + \frac{10}{100} \right)^{30} \quad \text{1 mark} \\
 &= 500 (1.1)^{30} \\
 &= \underline{\$ 8725} \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 (ii) A &= 500 (1.1^{30} + 1.1^{29} + \dots + 1.1^1) \\
 &= 500 (1.1^1 + 1.1^2 + \dots + 1.1^{30})
 \end{aligned}$$

$$\begin{aligned}
 \text{1 mark} \quad & \begin{array}{l} n = 30 \\ r = 1.1 \\ a = 1.1 \end{array} \quad \text{G.P.} \\
 S_n &= \frac{a(r^n - 1)}{r - 1} \\
 S_{30} &= \frac{1.1(1.1^{30} - 1)}{1.1 - 1} \\
 &= \frac{1.1(1.1^{30} - 1)}{0.1}
 \end{aligned}$$

$$\therefore A = \frac{500 \times 1.1 (1.1^{30} - 1)}{0.1}$$

$$= \underline{\$ 90472} \quad (2)$$

$$(d) 2x^2 + 4x - 7 = 0$$

$$\therefore x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-4 \pm \sqrt{4^2 - 4(2)(-7)}}{2(2)}$$

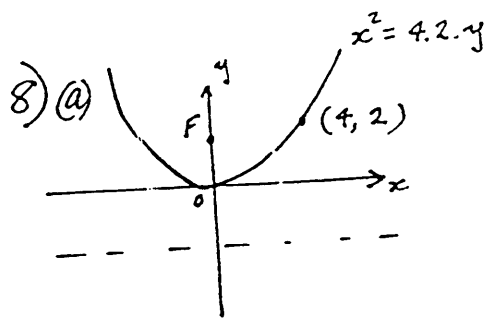
$$= \frac{-4 \pm \sqrt{16 + 56}}{4}$$

$$= \frac{-4 \pm \sqrt{72}}{4}$$

$$= \frac{-4 \pm 6\sqrt{2}}{4}$$

$$= \frac{2(-2 \pm 3\sqrt{2})}{2} \quad (2)$$

$$= \underline{-2 \pm 3\sqrt{2}}$$



(i)  $V(0,0)$  ①

(ii)  $F(0,2)$  ①

(iii) Dir.  $y=-2$  ①

(iv)  $y = \frac{x^2}{8}$

$$\therefore \frac{dy}{dx} = \frac{2x}{8}$$

$$= \frac{x}{4}$$

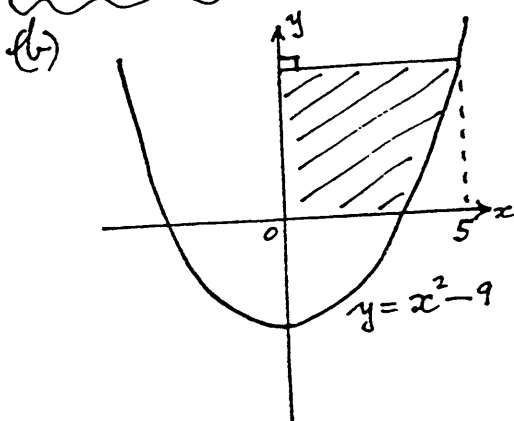
At  $x=4$   $f(4) = \frac{4}{4}$   
 $= 1 \leftarrow \text{mark}$

$\therefore$  eqn of tang. is

$$(y-2) = 1(x-4)$$

$$\therefore y-2 = x-4$$

$$\therefore x-y-2=0 \quad \textcircled{2}$$



$$x=5 \Rightarrow y=16$$

$$V = \int \pi x^2 dy \quad \text{1 mark}$$

$$= \pi \int_0^{16} (y+9) dy$$

$$= \pi \left[ \frac{y^2}{2} + 9y \right]_0^{16}$$

$$= \pi \{ (128 + 144) - 0 \}$$

$$= \underline{272\pi} \text{ units}^3 \quad \textcircled{2}$$

(c)  $y = e^{x^2}$

$$\frac{dy}{dx} = e^{x^2} \times 2x$$

$$= \underline{2xe^{x^2}} \leftarrow \text{mark}$$

$$\int_0^1 x e^{x^2} dx$$

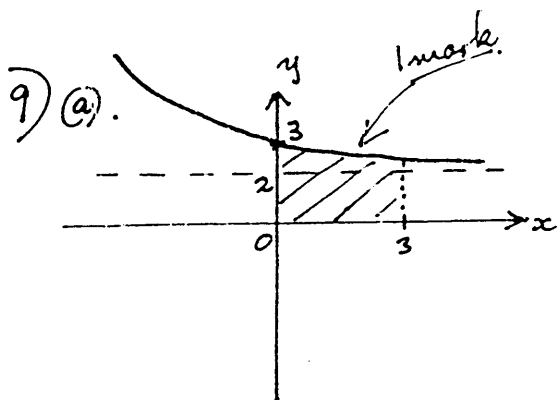
$$= \frac{1}{2} \int_0^1 2x e^{x^2} dx$$

$$= \left[ \frac{1}{2} \cdot e^{x^2} \right]_0^1 \leftarrow \text{mark}$$

$$= \frac{1}{2} [e^{x^2}]_0^1$$

$$= \frac{1}{2} \{ e^1 - e^0 \}$$

$$= \underline{\frac{1}{2}(e-1)} \quad \textcircled{3}$$



$$\begin{aligned}
 A &= \int_0^3 (e^{-x} + 2) dx \\
 &= [-e^{-x} + 2x]_0^3 \\
 &= (-e^{-3} + 6) - (-e^{-0} + 0) \\
 &= -\frac{1}{e^3} + 6 + 1 \\
 &= 7 - \frac{1}{e^3} \text{ units}^2 \quad (2)
 \end{aligned}$$

(b) (i)  $y = x^3 + 3x^2 - 9x$

$$\frac{dy}{dx} = 3x^2 + 6x - 9$$

For st. pts  $\frac{dy}{dx} = 0$

$$\therefore 3x^2 + 6x - 9 = 0$$

$$\therefore 3(x^2 + 2x - 3) = 0$$

$$(x+3)(x-1) = 0$$

$$\therefore x = 1, -3$$

At  $x=1$   $f''(x) = 6x + 6$   
 $= 6 + 6$   
 $> 0$

$\therefore$  MIN at  $(1, -5)$

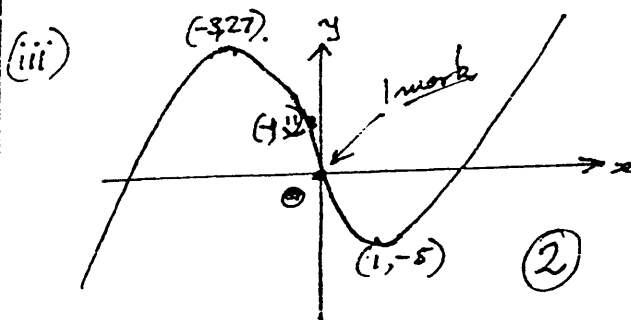
At  $x=-3$   $f''(-3) = -18 + 6$   
 $< 0$   
 $\therefore$  MAX at  $(-3, 27)$  (2)

(ii) Pts of inf  $f''(x) = 0$

$$\therefore 6x + 6 = 0$$

$$x = -1 \quad (1 \text{ mark})$$

$f''(-3) < 0$   
 $f''(1) > 0$  }  $\therefore$  sign change  
 $\therefore$  Pt of inf at  $(-1, 11)$  (2)



(iv) for mon. increasing  $\frac{dy}{dx} > 0$

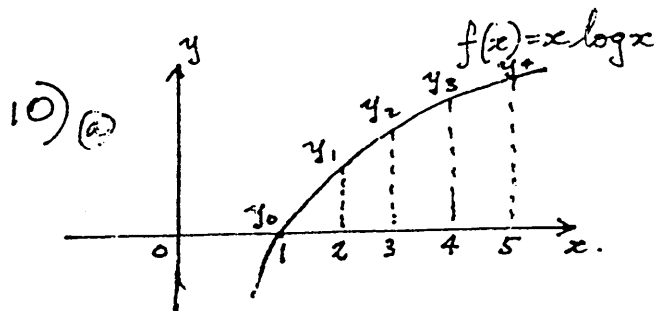
$$\therefore 3x^2 + 6x - 9 > 0 \quad (1 \text{ mark})$$

$$\therefore x^2 + 2x - 3 > 0$$

$$(x+3)(x-1) > 0$$



$$\therefore x < -3, x > 1 \quad (2)$$



$$\int_1^5 x \log x \, dx$$

formula

$$\div \frac{h}{3} \{ (y_0 + y_n) + 2 \text{ evens} + 4 \text{ odds} \}$$

$$= \frac{1}{3} \{ (f(1) + f(5)) + 2f(3) + 4(f(2) + f(4)) \}$$

subst.

$$= \frac{1}{3} \{ (0 + 8.05) + 2 \times 3.30 + 4(1.39 + 5.55) \}$$

$$= \frac{1}{3} (8.05 + 6.6 + 27.76)$$

there are alternate methods. (3)

$$= \underline{\underline{14.14}}$$

(b)

(i)  $x + x + h + x + h = 16$

$\therefore 2h + 3x = 16$

$\therefore 2h = 16 - 3x$

$\therefore h = 8 - \frac{3}{2}x$  (2)

(ii)

$A = \frac{1}{2} x \cdot \frac{\sqrt{3}}{2} x$  (triangle)

$+ xh$  (rectangle)

(mark)

$$= \frac{x^2 \sqrt{3}}{4} + x(8 - \frac{3}{2}x)$$

$$\therefore A = \frac{x^2 \sqrt{3}}{4} + 8x - \frac{3x^2}{2}$$

(2)

$$= 8x + \frac{x^2}{4} (\sqrt{3} - 6)$$

(iii)  $\frac{dA}{dx} = 8 + \frac{1}{2} x (\sqrt{3} - 6)$

$= 0$  when  $8 + \frac{x}{2} (\sqrt{3} - 6) = 0$

$\therefore (\sqrt{3} - 6)x = -16$

work  $x = \frac{-16}{(\sqrt{3} - 6)}$

$= \underline{\underline{3.75}}$

$\frac{d^2A}{dx^2} = \frac{\sqrt{3} - 6}{2}$

$< 0$

$\therefore \text{MAX.}$

Now  $h = 8 - \frac{3}{2}x$

$= 8 - \frac{3}{2} \times 3.75$

$= \underline{\underline{2.38}}$

$\therefore$  Dimensions of window for max are  $h = \underline{\underline{2.38 \text{ m}}}$  (3)

$x = \underline{\underline{3.75 \text{ m}}}$

[10]